Caucasus University

Caucasus School of Technology

Syllabus

Course Title	Python Programming
Course code	PYPR 2142.32
Status	Mandatory
ECTS	5.00
Level of study	Bachelor
Teaching semester	3

#	Lecturer	Working Place	Academic Degree	Academic position at Caucasus University	CU E-mail
1	Maksim Iavich	-		Professor	miavich@cu.edu.ge

Consultancy Time - Monday, Wednesday; 19.00-21.00

Format of the Course	
Lecture	26
Midterm/Final Eaxm	4.00
Out-of class preparation	125
Consultations	6

Objectives of the Course	• This course builds on Python Programming I and takes students from language fundamentals to building real, useful software. Students master Python as a multi-paradigm language — procedural, imperative, functional and object-oriented — and then apply it to two of the fastest-growing areas in tech: cybersecurity and applied artificial intelligence. • Rather than learning syntax in isolation, every core topic is paired with a modern, widely used Python library and a practical security or automation scenario. By the end of the semester students will have written automation scripts, a network reconnaissance tool, and a machine-learning-based intrusion detector — culminating in two portfolio-ready AI projects. •

Learning Outcomes

- On successful completion of the course, students will be able to:
- ● Design and reason about object-oriented, multi-paradigm programs;
- ● Select and apply appropriate data-structure types (tuples, lists, dictionaries, sets) for a given problem;
- ● Define, compose and test reusable functions and modules;
- ● Build practical software tools from third-party libraries and the pip ecosystem;
- ● Apply abstract thinking, analysis and synthesis to decompose problems into technical requirements;
- ● Write robust code with exception handling, assertions and clean documentation;
- ● Automate repetitive tasks and interact with the operating system, files, the web and external APIs;
- ● Use core cybersecurity libraries responsibly (network scanning, hashing, encryption) within legal and ethical boundaries;
- ● Integrate AI/ML models and Large Language Model (LLM) APIs into their own applications;
- ● Design, implement, document and defend a complete AI-based project.
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Compulsory Reading	Learning Python, 5th Edition O'Reilly Media Mark Lutz 2013

Supplementary Literature and Other Sources of Information	

Teaching Methods	1 Verbal or oral methods. 2 Working on books. 3 Writing. 4 Laboratory or visual demonstration methods 5 Practical methods 6 Discussions/debates 7 Team (Collaborative) work 8 Brain storming 9 Role play 10 Method of Demonstration 11 Induction, Deduction, Analysis, Synthesis. 12 Explanatory method 13 Action-oriented teaching

Evaluation System

Evaluation System

The aim of the evaluation is to assess to what extent the learning outcomes prescribed by the syllabus are reached. The student’s evaluation consists of multiple components and evaluates the course goals and learning outcomes by applying measurable criteria and appropriate rubrics. The student’s evaluation is based on four major principles: objectivity, trustworthiness, validity and transparency.

The students are evaluated according to two sets of evaluation: summative and formative. The aim of the summative assessment is to accurately evaluate the student’s performance. It monitors quality of learning and the level of the student’s achievement in relation to the goals set by the course. The formative assessment is oriented on the student’s development. It gives students appropriate feedback on their achievements.

The evaluation system includes 100 points and envisages:

a) Five types of positive grades:

a.a) (A) Excellent-91-100 points of assessment;

a.b) (B) Very good-81-90 points of maximal assessment;

a.c) (C) Good-71-80 points of maximal assessment;

a.d) (D) Satisfactory-61-70 points of maximal assessment;

a.e) (E) Sufficient-51-60 points of maximal assessment;

b) two negative grades:

b.a) (FX) Did not pass-41-50 points of maximal assessment, which means the student needs to work harder and is allowed to retake the exam one more time after working independently;

b.b) (F) Fail-40 points or less of maximal assessment, which means the students work is insufficient and he/she has to retake the course.

Students are awarded credits based on the final evaluation comprising the scores of the interim and final exam assessments.

The attainment of student’s learning outcomes considers the interim and final evaluations, for which relative proportions out of the total score (100 points) and a minimum competence level are allocated. Namely, out of 100 points, the interim results are allocated 70 points, while the final exam results are 30 points. In interim evaluations the minimum competency barrier to be reached is 59%. The interim evaluation includes assessment methods, the total of which is 70 points. For each assessment method, the evaluation is based on the pre-determined learning goals, task-oriented clear criteria and the learning rubrics drawn on their basis. In the interim results the student has to accumulate at least 59% of the 70 points to be allowed to take the final exam. The student’s final examination is passed, if he/she gets at least 60% of the total 30 points.

In case the student fails to overcome the minimum competency barrier of the final exam, he/she is allowed to retake the final examination. The student shall retake the final exam within the period prescribed by the academic calendar no later than 5 days after announcement of the results of the final exam.

In case the student totally scores 0-50 points or fails to overcome the minimum competency barrier set for any component of the evaluation (Interim/Final exam), he/she shall be given a grade of "F-0".

Knowledge Assessment Forms and Criteria				
Form of the Exam	**Quantity**	**Drop Quantity**	**Assessment**	**Total Points**
Final examination	1	0	30.00	30.00
Written work(Drop method)	3	1	10.00	20.00
Midterm Examination	1	0	25.00	25.00
Project / Report	2	0	12.50	25.00
Bonus	1	0	2.00	0.00
				100.00

Ways of Knowledge Evaluation	**Evaluation Criteria**
Final examination	Each task has a different weight. Evaluation per task: Correctness (40-50%): Produces correct output, meets requirements. Efficiency (10-20%): Uses optimal logic, avoids redundancy. Readability (10-15%): Well-structured, proper formatting. Error Handling (10-15%): Manages unexpected inputs, avoids crashes. Documentation (5-10%): Includes comments, explanations.
Written work(Drop method)	Each task has a different weight. Evaluation per task: Correctness (40-50%): Produces correct output, meets requirements. Efficiency (10-20%): Uses optimal logic, avoids redundancy. Readability (10-15%): Well-structured, proper formatting. Error Handling (10-15%): Manages unexpected inputs, avoids crashes. Documentation (5-10%): Includes comments, explanations.
Midterm Examination	Each task has a different weight. Evaluation per task: Correctness (40-50%): Produces correct output, meets requirements. Efficiency (10-20%): Uses optimal logic, avoids redundancy. Readability (10-15%): Well-structured, proper formatting. Error Handling (10-15%): Manages unexpected inputs, avoids crashes. Documentation (5-10%): Includes comments, explanations.
Project / Report	The students will have to create a project that reflects a type of project they could realistically be expected to complete in a real-life workplace. Students are allowed to use AI as a tool to support their work. Evaluation Criteria Each task has a different weight. Students will be evaluated on the following criteria: Correctness (40–50%): Produces correct output and meets all requirements. Efficiency (10–20%): Uses appropriate and efficient logic while avoiding unnecessary or redundant work. Readability (10–15%): Code/work is well-structured, clear, and properly formatted. Error Handling (10–15%): Handles unexpected inputs appropriately and avoids crashes or failures. Documentation (5–10%): Includes clear comments and explanations where necessary.
Bonus	The students will get the bonus for the activities at the conferences organized by cyber security centre at CU.

Academic Calendar	
I week	Lecture 2.00 hour
II week	Lecture 2.00 hour
III week	Lecture 2.00 hour
IV week	Lecture 2.00 hour
V week	Lecture 2.00 hour
VI week	Lecture/seminar 2.00 hour Written work(Drop method)
VII - IX week	Midterm Eaxm 2.00 hour
X week	Lecture 2.00 hour
XI week	Lecture/seminar 2.00 hour Project / Report
XII week	Lecture 2.00 hour
XIII week	Lecture/seminar 2.00 hour Written work(Drop method)
XIV week	Lecture 2.00 hour
XV week	Lecture/seminar 2.00 hour Project / Report
XVI week	Lecture/seminar 2.00 hour Bonus Written work(Drop method)
XVII - XIX week	Final Eaxm2.00 hour
XX week	Retake of Final Exam 2.00 hour

Course Overview	
Date	**Topics of Discussion**
Date 07/09/2026 **Hour** 15:45-17:40 **Auditorium** B14	**Topic** 1 Loops and Automation **Topics of Discussion** ● Algorithms with for ● Algorithms with while ● Automating repetitive tasks; iterating over IP ranges target lists **Compulsory Reading** O'Reilly Media Mark Lutz Learning Python, 5th Edition
Date 14/09/2026 **Hour** 15:45-17:40 **Auditorium** B14	**Topic** 2 Strings, Encoding and Hashing **Topics of Discussion** ● Complex tasks on strings and substrings ● Regular expressions (re) for parsing logs ● Encoding, decoding and integrity checks **Compulsory Reading** O'Reilly Media Mark Lutz Learning Python, 5th Edition 'Think Python 2nd Edition, by Allen B. Downey
Date 21/09/2026 **Hour** 15:45-17:40 **Auditorium** B14	**Topic** 3 Approximation and Search Algorithms **Topics of Discussion** ● Negligible-error approximation ● Binary search ● The Newton–Raphson method **Compulsory Reading** O'Reilly Media Mark Lutz Learning Python, 5th Edition
Date 28/09/2026 **Hour** 15:45-17:40 **Auditorium** B14	**Topic** 4 Functions and Modular Tooling **Topics of Discussion** Functions Modular Tooling ● Algorithms on functions ● Scope, arguments, return values ● Composing reusable security utilities **Compulsory Reading** O'Reilly Media Mark Lutz Learning Python; 'Think Python 2nd Edition, by Allen B. Downey
Date 05/10/2026 **Hour** 15:45-17:40 **Auditorium** B14	**Topic** 5 Recursion and Traversal **Topics of Discussion** ● Iterative vs. recursive algorithms; inductive reasoning ● Factorial, Towers of Hanoi, Fibonacci ● Recursive directory / web traversal **Compulsory Reading** O'Reilly Media Mark Lutz Learning Python;

Course Overview	
Date	**Topics of Discussion**
Date 12/10/2026 **Hour** 15:45-17:40 **Auditorium** B14	**Topic** 6 Files, Modules and the pip Ecosystem **Topics of Discussion** ● Working with files and modules ● Packages, virtual environments, pip ● Reading and writing logs reports **Compulsory Reading** O'Reilly Media Mark Lutz Learning Python; Written work(Drop method)
Midterm Eaxm, 2.00 Hour	
Date 09/11/2026 **Hour** 15:45-17:40 **Auditorium** B14	**Topic** 7 Tuples and Immutable Records **Topics of Discussion** Tuples Immutable Records ● Algorithms using tuples ● Immutable data for trusted records ● Coordinates, packets and fixed structures **Compulsory Reading** O'Reilly Media Mark Lutz Learning Python;
Date 16/11/2026 **Hour** 15:45-17:40 **Auditorium** B14	**Topic** 8 Lists and Data Handling **Topics of Discussion** Lists Data Handling ● Algorithms on lists ● Lists and functions ● Collecting hosts, ports and packets **Compulsory Reading** O'Reilly Media Mark Lutz Learning Python; Project / Report
Date 23/11/2026 **Hour** 15:45-17:40 **Auditorium** B14	**Topic** 9 Dictionaries, JSON and APIs **Topics of Discussion** ● Complex tasks on dictionaries ● Using dictionaries for efficiency ● JSON, REST APIs and mapping IP → service **Compulsory Reading** O'Reilly Media Mark Lutz Learning Python;

Course Overview	
Date	**Topics of Discussion**
Date 30/11/2026 **Hour** 15:45-17:40 **Auditorium** B14	**Topic** 10 Exception Handling and Robust Tools **Topics of Discussion** ● try/except blocks; managing errors and their causes ● The Exception class and custom subclasses ● Assertions **Compulsory Reading** O'Reilly Media Mark Lutz Learning Python; Written work(Drop method)
Date 07/12/2026 **Hour** 15:45-17:40 **Auditorium** B14	**Topic** 11 Sets, Wordlists and Deduplication **Topics of Discussion** ● Set theory ● Sets in practice ● Comparing two scans with set operations **Compulsory Reading** O'Reilly Media Mark Lutz Learning Python;
Date 14/12/2026 **Hour** 15:45-17:40 **Auditorium** B14	**Topic** 12 Classes and OOP Foundations **Topics of Discussion** ● Introduction and class examples (some material in Georgian) ● Scope/environment for classes; adding methods ● Set-of-integers class example **Compulsory Reading** O'Reilly Media Mark Lutz Learning Python; Project / Report
Date 21/12/2026 **Hour** 15:45-17:40 **Auditorium** B14	**Topic** 13 Inheritance and Building a Toolkit **Topics of Discussion** ● Inheritance ● Method overloading ● Method overriding; extensible tool frameworks **Compulsory Reading** O'Reilly Media Mark Lutz Learning Python; 'Think Python 2nd Edition, by Allen B. Downey Bonus Written work(Drop method)
Final Eaxm, 2.00 Hour	